

Problem-Solution Fit Canvas

Date	04 July 2026
Team ID	SOLO-CCFD-2026
Project Name	Credit Card Fraud Detection using Machine Learning
Maximum Marks	5 Marks

Canvas mapping the root causes, behaviors, and limitations of the target audience to the proposed ML-based solution.

CUSTOMER SEGMENT(S) Bank fraud/compliance analysts; Card-issuing banks; cardholders Indirectly protected	2. PROBLEMS / PAINS Manual review doesn't scale; fraud is rare (0.17%) and hidden among 284k+ transactions; delayed detection	3. TRIGGERS TO ACT A spike in chargebacks, a compliance audit, or customer complaints about unauthorized charges	4. EMOTIONS BEFORE/AFTER Before: anxious, overwhelmed. After: confident, reassured by automated screening
CUSTOMER LIMITATIONS Limited staff/time for manual review; CA-anonymized features are hard to interpret manually	9. PROBLEM ROOT CAUSE No automated, real-time, learning-based screening layer between transaction and settlement	7. BEHAVIOR + INTENSITY Analysts triage alerts reactively; intensity is high during fraud spikes	8. CHANNELS OF BEHAVIOR Online: fraud dashboards, email alerts. Offline: phone verification call to cardholders
AVAILABLE SOLUTIONS (Pros/Cons) Rule-based systems (fast, but rigid); manual review (accurate, but slow); our ML model (fast + adaptive, needs good data)			